Aware Mahalanobis Convolution and Polygon Structure Enhancement-Based Deep Network for Building Detection in Remote Sensing Images

Tong Gao, *Member, IEEE*, Yipeng Rao, Bin Bai, Xiaoning Hu, Zhuo Chen, Xiaorui Song, and Zhongke Yin

***Abstract*—As fundamental elements of urban landscapes, accurate building detection from high-resolution remote sensing imagery is an essential application of remote sensing information interpretation. However, this task remains challenging due to significant variations in building orientation, scale, and geometric shape. Existing methods rely on fixed kernel-based convolutions, struggle to adapt to diverse spatial scales, and fail to effectively model the intrinsic structures of buildings. To address these challenges, we propose aware Mahalanobis convolution (AMC)-polygon structure enhancement (PSE)-Net, an end-to-end building detection network. To better capture buildings with diverse orientations and sizes, the AMC module is presented, which utilizes a content-aware Mahalanobis transformation to automatically adjust the convolution kernels in both the orientation and scale domains based on image content, thereby effectively capturing architectural features across multiple directions and scales. To characterize the intrinsic geometric priors of regular building shapes, the PSE module is then developed, which integrates a self-supervised collinearity-preserving loss to inherit geometric priors and an Earth Mover's Distance (EMD)-based vertex regression loss to enhance consistency of detection results. In addition, to effectively mitigate semantic confusion between buildings and visually similar ground objects, such as roads and parking areas, the hierarchical spatial–semantic fusion (HSSF) head is designed to integrate multiple levels of semantic features to achieve robust detection. Experiments on the WHU dataset, the Inria Aerial Image Labeling dataset, and the Massachusetts Building dataset demonstrate that AMC-PSE-Net achieves 91.95% intersection over union (IoU) and 95.81% *F*1-score on the WHU dataset, outperforming state-of-the-art methods.**

***Index Terms*—Aware Mahalanobis convolution (AMC), building detection, polygon structure enhancement (PSE), self-supervised learning.**

I. Introduction

BUILDINGS are a typical landscape type in urban remote sensing imagery [1]. Compared with other object detection tasks, building detection is particularly intractable due to the diverse structural scales and varying orientations of buildings [2]. Consequently, developing a high-precision building detection method has emerged as a challenging research task that has received significant attention in recent years [3], [4].

Traditional building detection methods primarily rely on manually designed features [5] and are combined with prior knowledge and a classifier. Since manually designed features are difficult to describe the deep semantic information, these methods only perform well in limited scenarios [6]. In contrast, deep learning methods significantly enhance the accuracy and robustness of building detection by automatically learning effective features [7]. For instance, the representative CNN-based methods build up local receptive fields to extract multilevel features through stacked convolutional layers to achieve substantial performance improvements in building detection, e.g., fully convolutional networks (FCNs) [8], U-Net [9], and DeepLabv3+ [10]. To further enhance the delineation of building boundaries, attention mechanisms are introduced into the building detection networks [11]. Guo et al. [12] integrate spatial and channel attention mechanisms into a U-Net architecture to mitigate the semantic gap between features at different levels, thereby improving the accuracy of building boundary extraction. To increase the model's sensitivity to spatial positional information, FEPA-Net [13] employs a positional attention mechanism to encode building coordinates, leading to enhanced building detection results. Moreover, to capture the effective features in a specific orientation, the horizontal and vertical convolution (HVConv) module is constructed, which employs elongated convolutional kernels, e.g., 1×3 for horizontal convolutions and 3×1 for vertical convolutions, to expand the receptive field along different directions [14]. Additionally, to represent buildings with diverse structural sizes, MSANet [15] introduces a multiscale attention feature extraction (MAFE) module and a multiscale gated attention fusion (MGAF) module to effectively capture multiscale features of buildings.

Besides CNN-based local feature learning methods, the nonlocal-based methods presenting long-distance information extraction capabilities have received increasing attention recently. To capture nonlocal critical information, Vision Transformer (ViT) [16] converts 2-D images into 1-D sequences and employs self-attention mechanisms to model long-range dependencies. Wang et al. [17] introduce the BuildFormer framework, utilizing a dual-path structure combining a global context path and a spatial-detail context path to

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excavate deep features of objects. To address the information loss challenge inherent in CNN-based methods, MSSDMPA-Net incorporates a dynamic attention map-guided mechanism to enhance feature flow and spatial–spectral learning, thereby improving building extraction results [18].

Beyond building-specific detection networks, recent hyperspectral image super-resolution studies have also explored unsupervised, model-informed, and cross-modality learning paradigms for enhancing spatial–spectral representation under complex imaging conditions. For example, an enhanced deep image prior has been introduced to exploit the intrinsic image prior of neural networks for unsupervised hyperspectral image super-resolution [19]. Model-informed multistage unsupervised networks further incorporate imaging degradation priors into progressive reconstruction, improving the physical interpretability of hyperspectral super-resolution [20]. In addition, X-shaped interactive autoencoders with cross-modality mutual learning have been developed to strengthen the interaction between hyperspectral and multispectral modalities for unsupervised spatial-spectral enhancement [21]. Although these studies focus on hyperspectral image super-resolution rather than building footprint detection, they demonstrate the importance of exploiting intrinsic image priors, spatial-spectral information, and cross-modality interactions for robust remote sensing representation learning. Motivated by these advances, this work further investigates geometry-aware feature adaptation and polygon-structure regularization for accurate building detection in high-resolution remote sensing images.

The above methods belong to pixel-based detection methods. To represent the building shape, polygon-based methods use vertex series to describe building geometry to better learn the shape information of buildings. For instance, PolyWorld [22] formulates building extraction as vertex detection and connection using graph neural networks with optimal transport to determine the building contour. TransBuilding [23] represents buildings as bidirectional vertex sequences with structured attention to learn building shape under different vertex orderings. HiT [24] introduces hierarchical vertex- and edge-level attention to enhance contour regularity. HiSup [25] exploits hierarchical supervision from vertices, line segments, and regional masks to learn reversible building masks for accurate polygonal mapping of buildings. HD-Net [26] decouples body and boundary features via a high-resolution architecture to improve building detection results. However, these polygon-based methods cannot adaptively determine the number and order of vertices to capture the intrinsic geometric information of buildings, resulting in jagged edges or a loss of detail in the buildings.

Both the existing pixel-based methods and polygon-based methods have their own problems. The former can hardly adapt to the diverse orientations and sizes of buildings and cannot capture their inherent shape structure. The latter often struggle to define the proper vertex number and order that balances geometric fidelity with topological flexibility, particularly for complex building structures [27]. To address these challenges, an aware Mahalanobis convolution (AMC)-polygon structure enhancement (PSE)-Net is proposed to detect buildings with different orientations, scales, and shapes in remote sensing images. It contains two main modules: the AMC module and the PSE module. The AMC module introduces a small number of learnable conditional parameters to predict rotation angles, anisotropic scale factors, and fusion coefficients, enabling content-adaptive Mahalanobis transformation for orientation- and scale-aware convolution. In addition, to capture the regular shape information of the building, the PSE module is presented using unequal vertex matching with Earth Mover's Distance (EMD) and self-supervised collinearity-preserving regularization to explicitly integrate polygonal geometric priors into network optimization. By integrating AMC and PSE, the proposed AMC-PSE-Net achieves effective building detection in remote sensing images with complex urban scenes. The main contributions of this work are summarized as follows.

1) We propose an AMC module to improve the modeling of buildings with arbitrary orientations and scale variations. Different from standard, dilated, and deformable convolutions that rely on fixed kernels or spatial offsets, AMC predicts a small set of geometry-related parameters, including rotation angles, anisotropic scale factors, and fusion coefficients, conditioned on the input features. These parameters are used to transform and fuse base kernels in the kernel space, enabling convolutional responses to better align with orientation- and scale-dependent building structures.
2) We propose a PSE module to introduce structural polygon priors into network optimization. PSE formulates contour refinement as an asymmetric optimal transport problem and adopts an EMD-based loss to align predicted and ground-truth vertex sets with unequal cardinalities, without requiring explicit one-to-one matching. In addition, a self-supervised collinearity-preserving regularization term is introduced to encourage smooth building boundaries while preserving sharp corners, thereby improving the geometric regularity of building detection results.
3) Extensive experiments on the WHU, Inria, and Massachusetts building datasets demonstrate the effectiveness and robustness of AMC-PSE-Net. By integrating orientation-scale adaptive feature modeling, spatial-semantic feature fusion, and polygon-structure-aware supervision, the proposed method consistently improves detection accuracy and boundary quality. On the WHU dataset, AMC-PSE-Net achieves 95.81% $F1$-score and 91.95% intersection over union (IoU), obtaining the best performance among the compared methods.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents the architecture of AMC-PSE-Net and the mathematical formulation of its core modules. Section IV reports experimental settings, results, and analyses. Section V concludes this article and discusses future research directions.

II. Related Work

To better introduce the motivation of the proposed method, this section reviews related studies on label-efficient remote sensing representation learning, deformable

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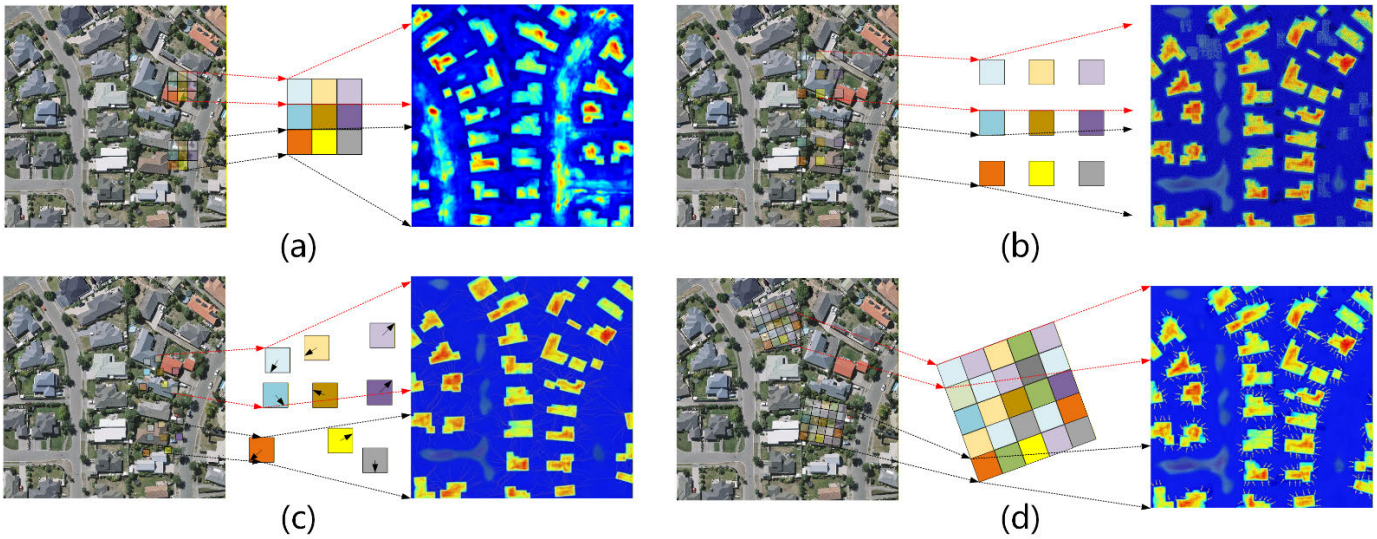


Fig. 1. Comparison of different convolutional feature mapping mechanisms. (a) Standard convolution uses a fixed local sampling pattern. (b) Dilated convolution enlarges the receptive field with a predefined dilation rate. (c) Deformable convolution learns unconstrained spatial offsets to increase sampling flexibility. (d) Proposed AMC, where rotation and anisotropic scale transformations are jointly predicted to align convolutional responses with buildings of different orientations and sizes.

convolution, dilated convolution, polygonal encoding, and recent remote sensing foundation models.

A. Geometry-Adaptive Convolution-Based Feature Learning

To better capture complex geometric variations in remote-sensing buildings, recent studies have augmented convolutional backbones with direction-aware attention mechanisms and deformable convolution operators. For example, CA-BASNet embeds coordinate attention into residual refinement blocks to enhance spatial awareness [28]. ADMNet introduces adaptive deformable convolution (ADC) to recover edge details while exploiting global context [29]. URDGNet [30] integrates an unpooling content reconstruction module with a directional fields guidance mechanism to predict pixel-level orientation fields for individual building instances, enabling geometrically precise boundary delineation.

Beyond offset-based deformation, explicit geometric adaptation of convolution kernels has also been shown to be effective for visual objects with strong orientation and scale variations. Pu et al. [31] proposed adaptive rotated convolution (ARC), in which convolution kernels are adaptively rotated according to object orientations, demonstrating that orientation-aware kernel transformation can enhance the representation of arbitrarily oriented objects. Zhang et al. [32] introduced scale-adaptive convolutions for scene parsing, where position-adaptive scale coefficients are learned to resize convolutional patches and generate flexible-size receptive fields. These studies indicate that kernel-space geometric adaptation is beneficial for improving feature representation when object appearances vary significantly in orientation or scale. However, existing rotation-aware and scale-aware designs usually model orientation and scale adaptation separately, while building detection in high-resolution remote sensing images requires joint modeling of orientation, anisotropic scale, and local structural regularity.

As illustrated in Fig. 1, the effectiveness of AMC feature mapping lies in its explicit geometry-aware kernel adaptation. Standard convolution uses a fixed square sampling pattern, which is insensitive to the dominant orientation of buildings. Deformable convolution introduces learnable offsets, but the offset distribution is weakly constrained and does not explicitly encode the orientation-scale structure of buildings. Inspired by orientation-aware kernel rotation [31] and scale-adaptive receptive field modeling [32], AMC further maps the base convolution kernels into a joint orientation- and scale-adaptive kernel space according to the input feature content. Specifically, AMC predicts the rotation angle, anisotropic scale factors, and fusion coefficients, so that the transformed kernels can align the effective receptive fields with elongated roofs, rotated building footprints, and multiscale structures. Compared with rotation-only or scale-only adaptation, the Mahalanobis formulation jointly models directional and anisotropic scale variations, making the feature responses more concentrated on building regions and less sensitive to background textures with similar appearances.

From a mathematical perspective, standard convolution performs feature aggregation using fixed and regularly spaced sampling locations

$$Y(p_0) = \sum_n W(p_n)X(p_0 + p_n) \tag{1}$$

where p_0 denotes the output location, p_n represents the predefined relative offsets of the convolution kernel, and W is a static kernel shared across all spatial positions.

Deformable convolution extends (1) by introducing learnable sampling offsets Δp_n

$$Y(p_0) = \sum_n W(p_n)X(p_0 + p_n + \Delta p_n) \tag{2}$$

with the offsets predicted in a data-driven manner as

$$\Delta p_n = f_{offset}(X). \tag{3}$$

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Although this mechanism increases sampling flexibility, the learned offsets lack explicit geometric or structural constraints, making the offset field unstable under strong orientation–scale coupling, which is common in dense urban scenes.

B. Dilated Convolution Based Feature Learning

To enlarge the effective receptive field and capture multiscale contextual information in remote-sensing building detection, dilated convolutions [33] are constructed by inserting dilation gaps, which aggregate broader spatial context without additional pooling operations, particularly beneficial for large buildings and scene-level reasoning.

Formally, a dilated convolution with dilation rates r is defined as

$$Y(p_0) = \sum_n W(p_n)X(p_0 + r \cdot p_n). \tag{4}$$

Despite their effectiveness in modeling scale variation, dilated convolutions follow fixed and isotropic sampling patterns, which cannot adaptively align their receptive fields with anisotropic building shapes exhibiting strong directional characteristics. When buildings present diverse rotations and scales, dilated convolution-based methods often fail to address them.

C. Polygonal Encoding-Based Feature Learning

Traditional pixel-based detection methods mainly optimize predictions at the pixel level and are insufficient to enforce global shape regularity. In comparison, polygonal modeling has become an important trend in high-quality building detection, as it provides a more compact and geometry-consistent representation of building structures [24].

In general, polygon-based methods represent a building as a set of vertices and a corresponding ground-truth building mask

$$P = \{v_i \in \mathbb{R}^2 \mid i = 1, 2, \ldots, N\}. \tag{5}$$

It learns to predict a corresponding set of vertices from the input image. Given an image I, the network outputs a set of predicted vertices

$$\hat{P} = \left\{\hat{v}_j\right\}_{j=1}^{M}, \quad \hat{v} = f_\theta(I)_j \tag{6}$$

where M is not necessarily equal to N, enabling flexible representation of building shapes. To supervise vertex prediction, many approaches formulate a dense vertex likelihood map and optimize it via a detection loss over the image domain.

To further ensure geometric consistency, the specific matching objective is introduced to align the predicted vertex set with the ground-truth polygon. This is commonly formulated as a set-to-set matching problem by minimizing the overall geometric discrepancy

$$L_{match} = \min_{\sigma} \sum_{i=1}^{N} \left\| \hat{V}_i - V_{\sigma(i)} \right\|_2 \tag{7}$$

where $\hat{V}_i$ and $\sigma(\cdot)$ denote the ith vertex of the predicted polygon and the cyclic shift function, respectively. This formulation enables order-preserving, one-to-one vertex matching with fixed cardinality and has been widely adopted in polygon-based detection frameworks for its simplicity and geometric clarity.

Note that existing polygon-based building detection methods still suffer from several critical limitations. Most approaches are restricted to single-topology polygon representations, making them unsuitable for buildings with complex topologies, such as those containing courtyards or holes, which often leads to incomplete or invalid reconstructions. In addition, the existing methods find it difficult to set the proper vertex number to balance geometric detail and shape regularity. Therefore, designing a polygon modeling framework that is topology-aware and vertex-adaptive remains a challenge in polygon-based building detection methods.

D. Label-Efficient Remote Sensing Representation Learning

Recent label-efficient remote sensing classification methods are closely related to robust feature representation under limited annotations. Lv et al. [34] proposed a novel sample augmentation approach for high-resolution remote sensing imagery, demonstrating that effective sample generation can improve classification performance when labeled samples are insufficient. Fu et al. [35] developed a CNN-Transformer and channel-spatial attention-based network for hyperspectral image classification with few samples, which jointly models local details, nonlocal dependencies, and channel-spatial feature interactions. Although these methods focus on remote sensing classification rather than building footprint detection, they indicate that sample augmentation, spatial–spectral representation, and attention-based feature interaction are important for improving model robustness under limited supervision.

E. Recent Remote Sensing Interpretation and Foundation Models

Recently, advanced remote sensing interpretation methods have increasingly explored new representation learning paradigms, including weakly supervised learning, Kolmogorov–Arnold Networks (KANs), and large foundation models. For example, SWCD [36] introduces similarity-aware weakly supervised learning for remote sensing change detection, where similarity information and large-kernel convolution attention are used to enhance feature continuity and multiscale representation. SSTKAN [37] further explores spectral–spatial–temporal KANs for hyperspectral change detection, demonstrating the potential of nonlinear representation learning for modeling complex remote sensing data distributions. In addition, remote sensing foundation models and vision-language models, such as RemoteCLIP [38] and SkySense [39], have shown strong generalization ability across diverse Earth observation tasks. AM-based methods have also been adapted to building extraction, such as SAMPolyBuild [40], which employs the segment anything model for polygonal building extraction. In addition, detection-guided SAM pipelines, such as YOLO-SAM [41], combine object detectors with SAM to generate building masks, providing another prompt-based solution for remote

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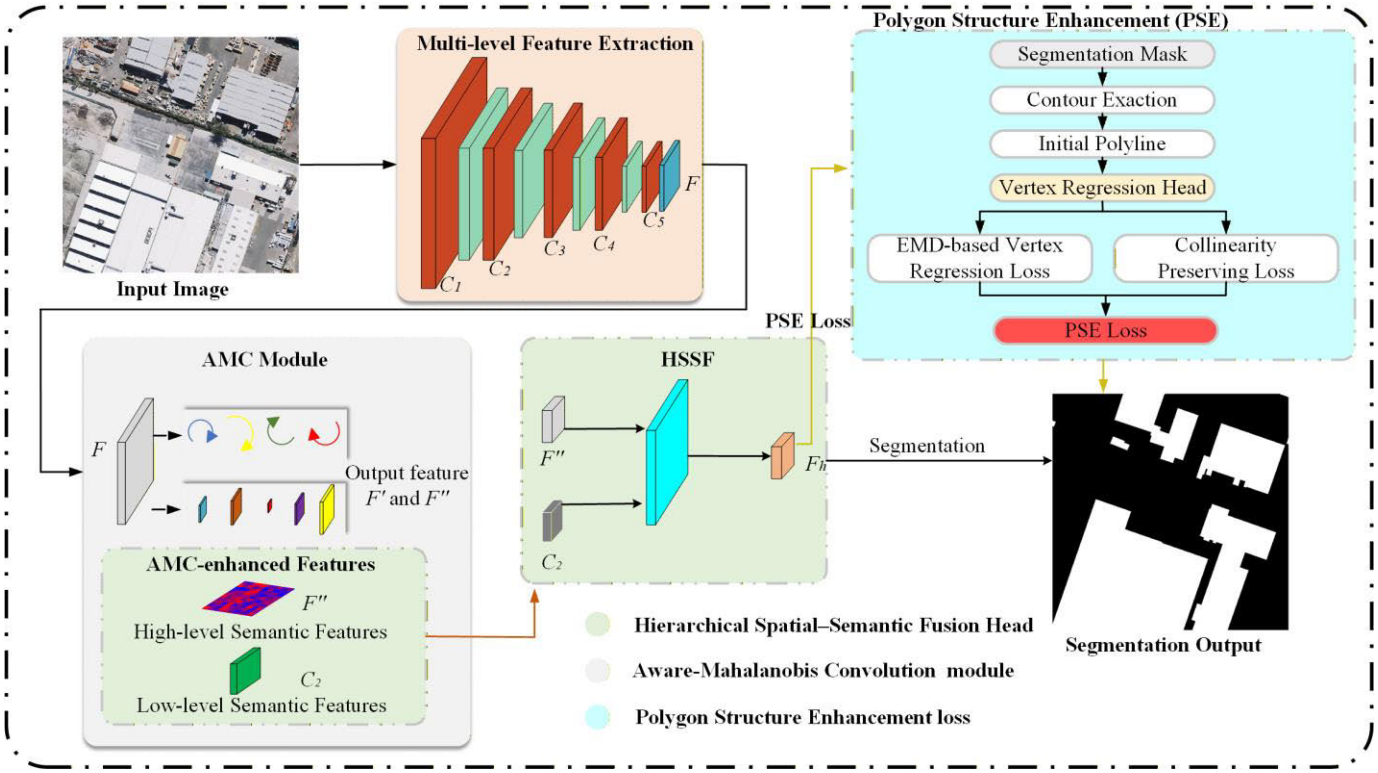


Fig. 2. Overall architecture of the proposed AMC-PSE-Net for building detection using remote sensing images.

sensing building extraction. These studies indicate that large-scale pretraining, weak supervision, and advanced nonlinear feature modeling are becoming important trends in remote sensing interpretation.

Although these methods focus on different tasks, such as change detection, hyperspectral interpretation, or foundation-model-based segmentation, they highlight the importance of robust multiscale representation, structural priors, and generalizable feature learning. Motivated by these trends, AMC-PSE-Net focuses on building detection by introducing content-adaptive Mahalanobis convolution and polygon-structure enhancement to improve orientation scale adaptability and geometric regularity.

III. Methodology

A. Overview

The overall architecture of the proposed AMC-PSE-Net is illustrated in Fig. 2. A VGG16 network is adopted as the backbone to extract multilevel semantic features.

To provide a controlled and reproducible baseline for the ablation study, RUNet is implemented as a U-shaped encoder–decoder segmentation network. The encoder adopts VGG16 to extract hierarchical feature representations, while the decoder progressively restores spatial resolution through upsampling and convolutional refinement. Skip connections are introduced between corresponding encoder and decoder stages to transfer low-level spatial details, which helps recover building boundaries and local structures. This configuration follows the widely used U-Net-based design in building footprint extraction and is consistent with MSA-UNet [42], a recent building extraction network that is built upon the U-Net framework and adopts VGG16 as the encoder for stronger feature representation. In addition, recent studies on building footprint extraction have also regarded U-Net-style encoder–decoder architectures as representative baseline frameworks, where the encoder is responsible for hierarchical feature extraction and the decoder is used for spatial-resolution recovery and pixel-level footprint prediction [43]. In this work, RUNet serves as the backbone baseline and does not include the proposed AMC, Hierarchical Spatial–Semantic Fusion (HSSF) Head, or PSE components. Therefore, under the same training protocol, the performance variations obtained by progressively introducing these modules can be attributed to the corresponding proposed designs rather than changes in the baseline architecture.

Specifically, given a remote sensing image X as input, the semantic feature extraction process of the AMC-PSE-Net backbone can be formulated as follows:

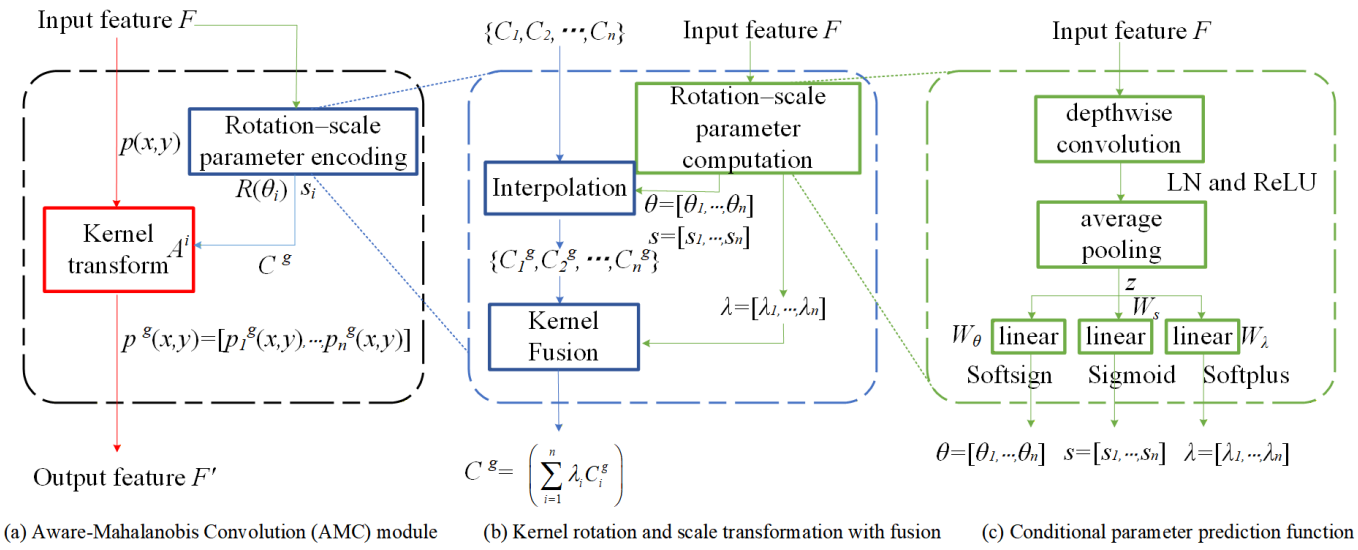
$$C_1, C_2, C_3, C_4, C_5 = VGG16(X \in \mathbb{R}^{3\times H\times W}). \tag{8}$$

Among the extracted features, the high-level semantic feature C_5 is further fed into an atrous spatial pyramid pooling (ASPP) module to capture rich contextual information and enlarge the effective receptive field

$$F = ASPP(C_5). \tag{9}$$

To handle arbitrary building orientations, scale variations, and irregular contours, AMC-PSE-Net integrates three complementary components. The AMC module performs content-aware kernel adaptation to capture orientation- and scale-sensitive building features, while the HSSF head combines high-level semantic representations with low-level

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(a) Aware-Mahalanobis Convolution (AMC) module (b) Kernel rotation and scale transformation with fusion (c) Conditional parameter prediction function

Fig. 3. Illustration of the proposed AMC module. (a) Overview of the AMC module. The adaptive convolution kernel is generated from the base kernel conditioned on the input feature map. (b) Kernel rotation-scale transformation and fusion. Multiple transformed kernels are generated via different rotation angles θ and scale parameters s, and then fused using content-dependent weights λ. (c) Conditional parameter prediction function. The input feature is encoded to predict θ, s, and λ in a data-dependent manner.

spatial details for improved boundary localization and reduced semantic confusion. In addition, the PSE module imposes polygon-level geometric constraints during training, encouraging more continuous, regular, and shape-consistent building boundaries. The network is jointly optimized with a pixel-wise segmentation loss and the PSE loss.

B. AMC Module

As illustrated in Figs. 1 and 3, conventional convolution and dilated convolution typically rely on static convolution kernels with fixed orientations and scales. Even deformable convolution only introduces learnable spatial offsets, while the kernels themselves remain constrained in their geometric configurations. Consequently, when buildings in remote sensing images exhibit arbitrary rotations and significant scale variations, conventional convolution still depends on direction- and scale-limited kernels, leading to suboptimal geometric alignment.

To address this limitation, the AMC module introduces a content-adaptive Mahalanobis transformation in the kernel space. This design is inspired by ARC [31] and scale-adaptive convolution [32], while further extending them to jointly model orientation and anisotropic scale variations in building footprints. Although ARC and scale-adaptive convolution can respectively alleviate orientation misalignment and scale variation, buildings usually exhibit more complex anisotropic structures, such as elongated rectangles, narrow roofs, and irregular polygons. Therefore, their long-axis and short-axis responses should be modeled jointly rather than independently.

AMC formulates this joint adaptation through a Mahalanobis-style metric in the kernel coordinate space. Specifically, only a compact set of content-dependent geometric parameters, including rotation angles and anisotropic scale factors, is predicted to describe the local orientation and structural scale of buildings. These parameters are used to transform the base kernels, enabling their effective receptive fields to align with the dominant direction and spatial extent of local building regions. Compared with directly increasing convolution parameters or learning dense spatial offsets, AMC represents complex orientation-scale variations with a small number of learnable geometric parameters, thereby improving the discriminative representation of buildings with arbitrary orientations, varying sizes, and anisotropic shapes.

Given the feature map F generated by the preceding ASPP module, AMC first predicts the rotation, scale, and fusion parameters through a conditional geometric-parameter prediction function, as shown in Fig. 3(c). Specifically, the input feature is processed by a depthwise convolution, layer normalization, ReLU activation, and global average pooling. The resulting global descriptor is then fed into three parallel linear heads to generate the rotation angles θ, anisotropic scale factors s, and content-dependent fusion coefficients λ

$$z = GAP(ReLU(LN(DWConv(F)))) \tag{10}$$

$$\theta = Softsign(W_\theta z), \quad s = Sigmoid(W_s z), \quad \lambda = Softplus(W_\lambda z) \tag{11}$$

where z denotes the context descriptor extracted from the input feature map. The predicted parameter groups are defined as

$$\theta = [\theta_1, \theta_2, \ldots, \theta_n], \quad s = [s_1, s_2, \ldots, s_n]$$
$$\lambda = [\lambda_1, \lambda_2, \ldots, \lambda_n]. \tag{12}$$

Here, n denotes the number of base kernels or transformation branches. For the ith branch, θ_i represents the rotation angle, $s_i = (s_{x,i}, s_{y,i})$ denotes the anisotropic scale factors along two orthogonal axes, and λ_i is the content-dependent fusion coefficient. The Softsign, Sigmoid, and Softplus functions are used to constrain the predicted parameters into stable ranges, which improves the robustness of the kernel transformation and fusion process.

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Let the set of base convolution kernels be denoted as

$$C = \{C_1, C_2, \ldots, C_n\} \tag{13}$$

where C_i denotes the ith base kernel.

For each base kernel C_i, AMC performs a rotation-scale transformation in the kernel coordinate space. Let

$$p(x, y) = [x, y]^T \tag{14}$$

denote a sampling coordinate centered at the kernel origin. The transformed coordinate $p_i^g(x, y)$ is defined as

$$p_i^g(x, y) = A_i p(x, y). \tag{15}$$

The transformed coordinate set of all branches can be written as $p^g(x, y) = [p^1 g(x, y), p^2 g(x, y), \ldots, p^n g(x, y)]$. Where A_i is the geometry transformation matrix determined by the predicted rotation angle and anisotropic scale factors

$$A_i = R(\theta_i) s_i \tag{16}$$

$$R(\theta_i) = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{bmatrix} \tag{17}$$

$$s_i = \begin{bmatrix} s_{x,i} & 0 \\ 0 & s_{y,i} \end{bmatrix}. \tag{18}$$

In this formulation, $R(\theta_i)$ controls the orientation of the kernel coordinate system, while s_i controls the anisotropic scaling along two orthogonal directions. Therefore, the transformed sampling coordinates are no longer restricted to a fixed square pattern, but become adaptive to the dominant orientation and structural scale of building regions. This rotation-scale transformation provides a Mahalanobis-style kernel mapping because it jointly models direction-dependent and axis-dependent variations in the kernel coordinate space. As a result, the transformed kernel can produce different responses along different structural axes, which is particularly suitable for anisotropic building structures such as elongated roofs and rotated building footprints.

Since the transformed coordinates $p_i^g(x, y)$ are generally continuous rather than discrete grid locations, AMC adopts interpolation-based kernel resampling to generate the transformed kernel C_i^g. For each coordinate p in the transformed kernel space, the corresponding kernel weight is obtained by sampling the original base kernel C_i at the inverse-transformed coordinate

$$C_i^g = I\left(C_i, A_i^{-1} p\right) \tag{19}$$

where $I(\cdot)$ denotes the interpolation operation in the kernel space. Through this resampling process, each base kernel C_i is mapped into a geometry-adaptive kernel C_i^g, whose orientation and anisotropic scale are determined by the input-dependent parameters θ_i and s_i.

Considering that a remote sensing scene may contain buildings with multiple dominant orientations and characteristic scales, AMC further fuses multiple transformed kernels using the predicted content-dependent coefficients. As shown in Fig. 3(b), the fused adaptive kernel C^g is obtained as

$$C^g = \sum_{i=1}^{n} \lambda_i C_i^g \tag{20}$$

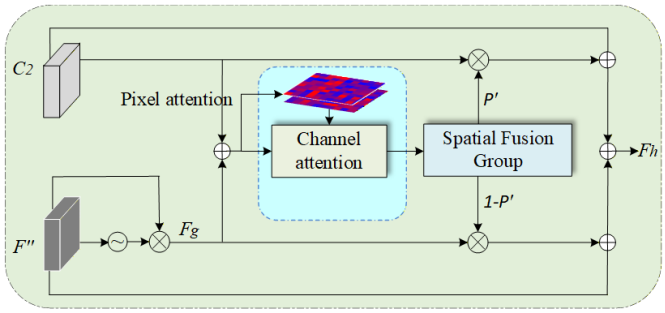


Fig. 4. Illustration of the HSSF head.

where λ_i denotes the fusion coefficient of the ith transformed kernel. A direct implementation can first convolve each transformed kernel with the input feature map F, and then aggregate the responses using the predicted coefficients

$$F' = \sum_{i=1}^{n} \lambda_i \left(C_i^g * F\right). \tag{21}$$

According to the linear property of the convolution operator, the above formulation can be equivalently rewritten as

$$F' = \left(\sum_{i=1}^{n} \lambda_i C_i^g\right) * F = C^g * F. \tag{22}$$

This indicates that AMC does not need to perform multiple independent convolutions during implementation. Instead, the transformed kernels are first fused into a single geometry-adaptive kernel C^g, and then one convolution operation is applied to obtain the output feature map F'. Therefore, AMC avoids the explicit execution of multiple independent convolution branches.

In summary, AMC predicts content-dependent geometric parameters from F to transform the base kernels $\{C_1, C_2, \ldots, C_n\}$ into geometry-adaptive kernels $\{C_1^g, C_2^g, \ldots, C_n^g\}$. These transformed kernels are then fused into C_g and applied to F to generate F', allowing the convolutional responses to adapt to buildings with arbitrary orientations, varying sizes, and anisotropic geometric structures.

The output feature F' represents the orientation- and scale-adaptive high-level feature generated by the first AMC operation. To further strengthen geometry-aware semantic representation, F' is subsequently fed into another AMC module to obtain the enhanced high-level semantic feature F''. This feature is then used as the semantic input of the HSSF module for subsequent spatial-semantic fusion with the low-level feature C_2.

C. HSSF Head

As illustrated in Fig. 4, to mitigate semantic confusion between buildings and visually similar surfaces, we propose an HSSF head for complementary integration of high-level semantic features and low-level spatial details. The high-level feature map provides strong category-level discrimination but lacks spatial precision, while the low-level map contains rich spatial information but weak semantics. To suppress redundant textures, the low-level feature is refined via gating; meanwhile,

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pixel-level attention encodes spatial importance. These are combined and modulated by a channel attention mechanism conditioned on spatial content, unlike conventional globally shared channel attention.

Specifically, C_2 and F'' denote the low-level spatial detail feature and the high-level semantic feature, respectively. To suppress redundant responses and enhance discriminative semantic information, F'' is first processed by a gating unit to generate the gate-enhanced semantic feature F_g. Then, C_2 and F_g are preliminarily fused to form an intermediate feature map containing both spatial details and semantic information. The fused feature is further enhanced sequentially by pixel attention and channel attention. The pixel attention mechanism highlights spatial responses around building regions and boundary locations, while the channel attention mechanism adaptively recalibrates the importance of different semantic channels.

The attention-enhanced feature is then fed into the spatial fusion group to generate the spatially adaptive fusion weight map P'. The spatial fusion group consists of a 3×3 convolution, batch normalization, ReLU activation, a 1×1 convolution, and a sigmoid function. Therefore, P' serves as a pixel-wise fusion weight to adaptively regulate the contributions of low-level spatial details and high-level semantic information. According to the fusion path shown in Fig. 4, a larger value of P' emphasizes the boundary structure and geometric details from C_2, whereas a smaller value relatively enhances the high-level semantic discrimination from Fg. The complementary fusion between high-level and low-level features is formulated as

$$P' = SpatialFusionGroup(ChannelAttention (PixAttention(C_2 \oplus \mathrm{F_g}))) \quad (23)$$

$$F_h = \sigma\left(P\prime \otimes C_2 + (1 - P\prime) \otimes F_g\right) \quad (24)$$

where $\sigma(\cdot)$ and $\otimes$ denote sigmoid function and element-wise multiplication, respectively. P' denotes the spatially adaptive fusion weight map generated by the spatial fusion group, and F_h is the predicted binary building mask. This formulation ensures that semantic information dominates in homogeneous regions, while spatial details are emphasized near building boundaries and structurally complex areas.

From the above formulation, it can be observed that HSSF performs location-dependent feature fusion rather than relying on a single global attention weight per channel. By dynamically balancing semantic dominance and structural fidelity through the spatially varying weight map P', the proposed fusion head effectively enhances building discrimination and boundary integrity, thereby alleviating semantic confusion between buildings and visually similar man-made surfaces in complex urban scenes.

D. Polygon Structure Enhancement

To further improve the geometric regularity of building detection results, the PSE module is introduced to incorporate polygon-level structural constraints into network optimization, as illustrated in Fig. 5. Specifically, given the predicted binary building mask F_h, PSE first extracts the building contour and converts it into an initial polyline representation. Then, a learnable vertex regression head refines the contour vertices. Finally, an EMD-based vertex regression loss and a self-supervised collinearity-preserving loss are jointly used to guide the predicted polygon toward a structurally regular and shape-preserving configuration.

1) Unequal Vertex Regression Loss: According to the predicted binary building mask F_{h}, we extract the outer boundary of each building instance using a contour tracing algorithm. To reduce the computational burden caused by excessive contour points, we uniformly resample the contour with a fixed step size s_2. The resulting polyline is defined as

$$Q = \{q_1, q_2, \ldots, q_T\}, \quad q_i = (x_i, y_i) \quad (25)$$

where Q denotes the initial contour polyline extracted from the predicted binary building mask F_h after contour extraction, T is the number of resampled contour points, and $q_i = (x_i, y_i)$ represents the coordinate of the ith contour point. The resampling step s_2 is used to control the density of contour points and reduce redundant boundary samples.

To construct a structured polygon representation, the polyline is partitioned into fixed-length segments of size K, forming a polyline tensor

$$Q \in R^{M \times K \times 2} \quad (26)$$

where K denotes the number of contour points in each segment, M denotes the number of polyline segments, and the reshaped representation of Q has the size $R^{M \times K \times 2}$. In this representation, the total number of contour points is $T = M \times K$ after padding or uniform resampling.

To supervise contour refinement and establish reliable alignment between predicted and ground-truth vertices, the initial contour polyline Q is refined by a learnable vertex regression head. Let F_p denote the feature map used for vertex regression. For each contour point in Q, grid sampling is performed on F_p to extract the aligned visual feature at the corresponding spatial location. The sampled visual feature is concatenated with positional encoding and then fed into a Transformer decoder to predict coordinate offsets, yielding the refined polyline tensor

$$\tilde{Q} = Q + \mathrm{Transformer}(\mathrm{GridSample}(F_p, Q)) \quad (27)$$

where $\tilde{Q}$ denotes the refined polyline tensor, F_p denotes the feature map used for vertex regression, GridSample$(\cdot)$ performs the bilinear interpolation on F_p according to the coordinates of Q, and the Transformer decoder predicts coordinate offsets for contour refinement.

The refined predicted vertex set is denoted as

$$\tilde{Q} = \{\tilde{q}_1, \tilde{q}_2, \ldots, \tilde{q}_{Tpred}\}. \quad (28)$$

The regression head incorporates position-sensitive attention to enhance perception of local geometric structures. The ground-truth vertex set

$$\bar{Q} = \{\bar{q}_1, \bar{q}_2, \ldots, \bar{q}_{Tgt}\} \quad (29)$$

where *Tpred* and *Tgt* denote the numbers of predicted and ground-truth vertices, respectively.

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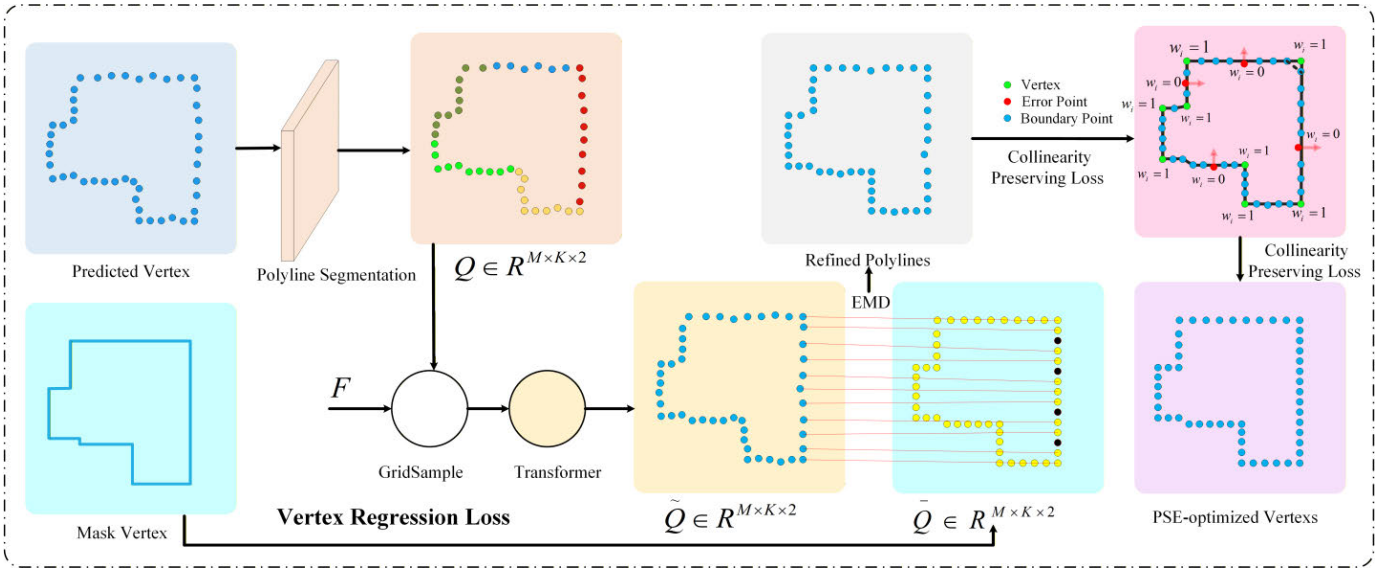


Fig. 5. Illustration of polygon vertex encoding loss for building detection.

Since the two sets may have different cardinalities, we treat them as two equally weighted discrete distributions

$$w_i^{pred} = \frac{1}{T_{pred}}, w_j^{gt} = \frac{1}{T_{gt}} \tag{30}$$

where w_i^{pred} and w_j^{gt} denote the normalized masses assigned to the ith predicted vertex and the jth ground-truth vertex, respectively.

We employ EMD to compute the optimal transport plan between the predicted vertex set $\tilde{Q}$ and the ground-truth vertex set $\bar{Q}$. The EMD-based vertex regression loss is defined as the minimum transportation cost

$$L_{vert} = \min_{f_{ij}\geq 0} \sum_{i=1}^{T_{pred}} \sum_{j=1}^{T_{gt}} f_{ij} \left\| \tilde{q}_i - q_j^- \right\|_2 \tag{31}$$

subject to the constraints

$$\sum_{j=1}^{T_{gt}} f_{ij} = w_i^{pred} \quad \forall i = 1, 2, \ldots, T_{pred} \tag{32}$$

$$\sum_{i=1}^{T_{pred}} f_{ij} = w_j^{gt} \quad \forall j = 1, 2, \ldots, T_{gt} \tag{33}$$

where f_{ij} denotes the transport mass from the ith predicted vertex $\tilde{q}_i$ to the jth ground-truth vertex $\bar{q}_j$. The first constraint ensures that the total mass transported from each predicted vertex equals its normalized mass, while the second constraint ensures that the total mass received by each ground-truth vertex equals its normalized mass. Therefore, the EMD loss can align predicted and ground-truth vertex sets with unequal cardinalities without requiring predefined vertex ordering or explicit one-to-one matching.

2) Self-Supervised Collinearity-Preserving Loss: Corners may remain. To address this, a unified self-supervised geometric loss is designed to suppress jagged edges along straight segments and preserve sharp corner structures. Importantly, this loss depends only on predicted vertices $\tilde{Q}$ and requires no additional supervision. For each interior vertex $\tilde{q}_i$, we define its perpendicular distance to the line formed by its neighbors

$$d_i = dist(\tilde{q}_i, line(\tilde{q}_{i-1}, \tilde{q}_{i+1})) \tag{34}$$

where d_i denotes the perpendicular distance from the interior vertex $\tilde{q}_i$ to the line connecting its two neighboring vertices $\tilde{q}_{i-1}$ and $\tilde{q}_{i+1}$. A smaller d_i indicates stronger local collinearity, while a larger d_i usually corresponds to a potential corner point.

To distinguish straight line points from corners, we define a monotonic confidence weight

$$w_i = \sigma(d_i - \mu) \tag{35}$$

where $\sigma(\cdot)$ is the Sigmoid function and μ is a threshold parameter. Thus, when $w_i \approx 0$, the vertex corresponds to a straight segment, and when $w_i \approx 1$, it corresponds to a corner point.

We combine both objectives into a single loss function

$$L_{geo} = \sum_{i=2}^{Tpred-1} [(1 - w_i)d_i - \alpha w_i d_i]. \tag{36}$$

It can be rewritten as

$$L_{geo} = \sum_{i=2}^{Tpred-1} [1 - (1 + \alpha)w_i]d_i. \tag{37}$$

For straight segments where $w_i \approx 0$, the coefficient is positive, and minimizing the geometric loss L_{geo} reduces d_i, thereby enforcing collinearity. For corner points where $w_i \approx 1$, the coefficient $1 - (1 + \alpha)w_i$ becomes negative when $\alpha > 0$. Therefore, minimizing L_{geo} encourages a larger d_i under the constraint of the EMD alignment loss, which helps preserve sharp corners. In our implementation, α is set to 1, which simplifies the coefficient to $1 - 2w_i$.

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(a) (b) (c)

Fig. 6. Representative images in the WHU-building dataset. (a) Contains buildings with different orientations. (b) Contains buildings with similar color to the road. (c) Contains buildings with small sizes.

IV. Experiments and Analysis

To comprehensively evaluate the performance advantages of the proposed AMC-PSE-Net over SOTA methods in building detection, three publicly available datasets, i.e., the WHU Building dataset, the Inria Aerial Image Labeling dataset, and the Massachusetts Building dataset, are applied to thoroughly examine different performance metrics for each method. The details of these three datasets are given as follows.

1) *WHU-Building Dataset:* The WHU Building dataset combines both satellite and aerial imagery [44]. In this article, the aerial images are selected for performance evaluation, which spans over 450 square kilometers and contains approximately 22 000 individual buildings. These buildings present different sizes, various directions, and diverse structures. The dataset includes a variety of building structures, including typical rectangular shapes, complex polygons, L-shaped, and T-shaped buildings. To meet the training requirements of different deep learning methods, these images are cropped to 512×512 pixels, generating 8188 images. To ensure the implementation of different methods, these images are scientifically divided into three subsets: the training set with 4736 images, the validation set with 1036 images, and the test set with 2416 images. The representative images in the WHU-Building dataset are displayed in Fig. 6.
2) *Inria Aerial Image Labeling Dataset:* The Inria Building Dataset [11] is released by the French National Institute for Research in Computer Science and Automation (INRIA). This dataset is widely used for building detection and semantic segmentation tasks, with a spatial resolution of 0.3 m. These images are captured from five different cities: Austin, Chicago, Gisap, Tyrol, and Vienna, and thus present different backgrounds. We select $1 \sim 5$ images per city for validation and the remainder for training. To meet the computational and memory requirements for training, the original 5000×5000 pixel images are cropped into 512×512 pixel patches. The cropped images that did not contain buildings were discarded. Finally, 9737 images are retained for training and 1942 images for validation. It covers a variety of backgrounds, including high-density urban buildings, low-density suburban residences, and mountainous towns.
3) *Massachusetts Building Dataset:* The Massachusetts Building Dataset [45] consists of 151 aerial images from the Boston area, each with a size of 1500×1500 pixels and a spatial resolution of 1 m. This dataset includes both urban and suburban scenes, where the buildings vary in size, shape, texture, and color. It presents significant challenges for building detection tasks and is well-suited for validating the effectiveness of detection networks. The dataset is divided into training, validation, and test sets, containing 137, 4, and 10 images, respectively. The dataset contains high-density buildings in different backgrounds, providing comprehensive verification for building detection methods. In our experiments, we followed the default dataset division and resized the images to 512×512 pixels. As a result, 3699 images are obtained for training, 36 images for validation, and 90 images for testing, respectively. Our code of AMC-PSE-Net is available at: https://github.com/supergt3/AMC-PSE-Net

A. Evaluation Metrics

To evaluate the performance of different methods, four widely used evaluation metrics, i.e., $F1$ score, IoU, precision, and recall [46], are introduced to measure the building detection results for the proposed AMC-PSE-Net and the comparison methods. The definitions of these metrics are shown below

$$IoU = \frac{N_{TP}}{N_{TP} + N_{FP} + N_{FN}} \tag{38}$$

$$Precision = \frac{N_{TP}}{N_{TP} + N_{FP}} \tag{39}$$

$$Recall = \frac{N_{TP}}{N_{TP} + N_{FN}} \tag{40}$$

$$F1 = 2 * \frac{Recall * Precision}{Recall + Precision} \tag{41}$$

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TABLE I
QUANTITATIVE ABLATION RESULTS ON WHU, INRIA, AND MASSACHUSETTS

Base	AMC	HSSF	PSE	WHU (%)				Inria (%)				Massachusetts (%)			
				F1↑	IoU↑	Pre↑	Recall↑	F1↑	IoU↑	Pre↑	Recall↑	F1↑	IoU↑	Pre↑	Recall↑
√		√	√	95.32	91.06	95.00	95.64	90.13	82.04	91.24	89.05	86.24	75.80	**88.34**	84.23
√	√		√	95.59	91.54	95.60	**95.57**	90.42	82.52	91.43	89.44	86.45	76.14	88.00	84.96
√	√	√		95.65	91.66	95.80	95.50	90.03	81.87	91.36	88.74	86.47	76.16	87.60	85.30
√				94.58	89.72	94.26	94.90	89.71	81.34	91.37	88.11	85.12	74.09	85.87	84.39
√	√	√	√	**95.81**	**91.95**	**96.10**	95.52	**90.75**	**83.07**	**92.01**	**89.52**	**86.64**	**76.44**	88.02	**85.32**

where N_{TP}, N_{FP}, N_{TN}, and N_{FN} denote the correctly predicted positive pixels (buildings), the incorrectly predicted positive pixels (nonbuildings), the correctly predicted negative pixels (nonbuildings), and the incorrectly predicted negative pixels (buildings), respectively.

B. Parameter Settings

All experiments are implemented using PyTorch 1.8.1 (CUDA 12.2) on a single NVIDIA RTX 3090 GPU, running on Ubuntu 18.04.

During training, the AdamW optimizer and a cosine learning rate scheduling strategy are used based on cyclical expansion with an initial cycle $T_0 = 15$. Across all three benchmark datasets, the models are trained for 105 epochs with an initial learning rate of 5e-4. Due to the high spatial resolution (512×512) of input images and the GPU memory constraint (24 GB on a single NVIDIA RTX 3090), the batch size is uniformly set to 4 for all datasets. During inference, horizontal and vertical flips are applied as part of our Test-Time Augmentation (TTA) strategy. The number of base kernels n in the AMC module is set to 6 in all experiments.

C. Loss Function

To jointly recover fine-grained building boundaries and capture long-range structural dependencies, we optimize the model using a hybrid loss function that integrates standard cross-entropy (CE) loss with two geometry-aware terms, the EMD-based vertex regression loss L_{vert} and the self-supervised collinearity-preserving loss L_{geo}. It is formulated as follows:

$$L_{total} = L_{CE} + \upsilon \cdot L_{vert} + \gamma \cdot L_{geo} \tag{42}$$

where υ and γ denote trade-off parameters used to balance the relative contributions of the individual loss components, both set to 1 in all experiments.

D. Ablation Study

1) Experimental Analysis: To validate the effectiveness of each component in the proposed AMC-PSE-Net, comprehensive ablation experiments are conducted on three public building detection datasets, namely WHU, Inria, and Massachusetts. For fair and standardized comparison, all ablation variants are constructed as controlled modifications of the same RUNet baseline. RUNet is selected because it follows a widely adopted U-shaped encoder–decoder architecture with VGG16 features and skip connections, which provides a stable and commonly used segmentation backbone for building extraction while avoiding task-specific geometric or attention designs. In all variants, the backbone, decoder configuration, input resolution, data split, optimizer, training schedule, and evaluation protocol are kept unchanged. The baseline does not include the proposed AMC, HSSF head, or PSE supervision, and is trained only with the pixel-wise CE loss. AMC, HSSF, and PSE are then introduced progressively under the identical training protocol. Therefore, the performance differences reported in Table I can be attributed to the added modules rather than to changes in the backbone, data processing, or training configuration. The quantitative results are reported in Table I, where Pre denotes precision, and the best results are highlighted in bold.

The results in Table I show that removing any single component leads to consistent performance degradation across all three datasets, indicating that each module contributes positively and that their effects are complementary rather than redundant. The most noticeable performance drop occurs when the AMC module is removed. In this case, the $F1$-score on the Inria dataset decreases from 90.75% to 90.13%, while the IoU drops from 83.07% to 82.04%. Similar trends are observed on the WHU and Massachusetts datasets, where the $F1$-score decreases by 0.49% and 0.40%, respectively, accompanied by corresponding reductions in IoU. The larger degradation on the Inria dataset can be attributed to its complex urban scenes, in which buildings exhibit large variations in orientation, scale, and spatial arrangement. These results indicate that AMC effectively enhances feature representation by adapting convolutional responses to building geometry, thereby improving discrimination under challenging conditions.

When the HSSF module is removed, the overall performance also degrades consistently across all datasets, with a particularly noticeable decline in precision. On the WHU dataset, precision decreases from 96.10% to 95.60%, and the $F1$-score drops accordingly, suggesting an increased number of false positives. This behavior is consistent with the role of HSSF in aligning high-level semantic information with low-level spatial details, which is essential for distinguishing buildings from visually similar man-made structures such as roads and parking lots. Comparable performance degradation is observed on the Inria and Massachusetts datasets, further confirming the effectiveness of HSSF in complex scenes.

When the PSE loss is removed, the $F1$-score and IoU on the WHU dataset decrease from 95.81% to 95.65% and

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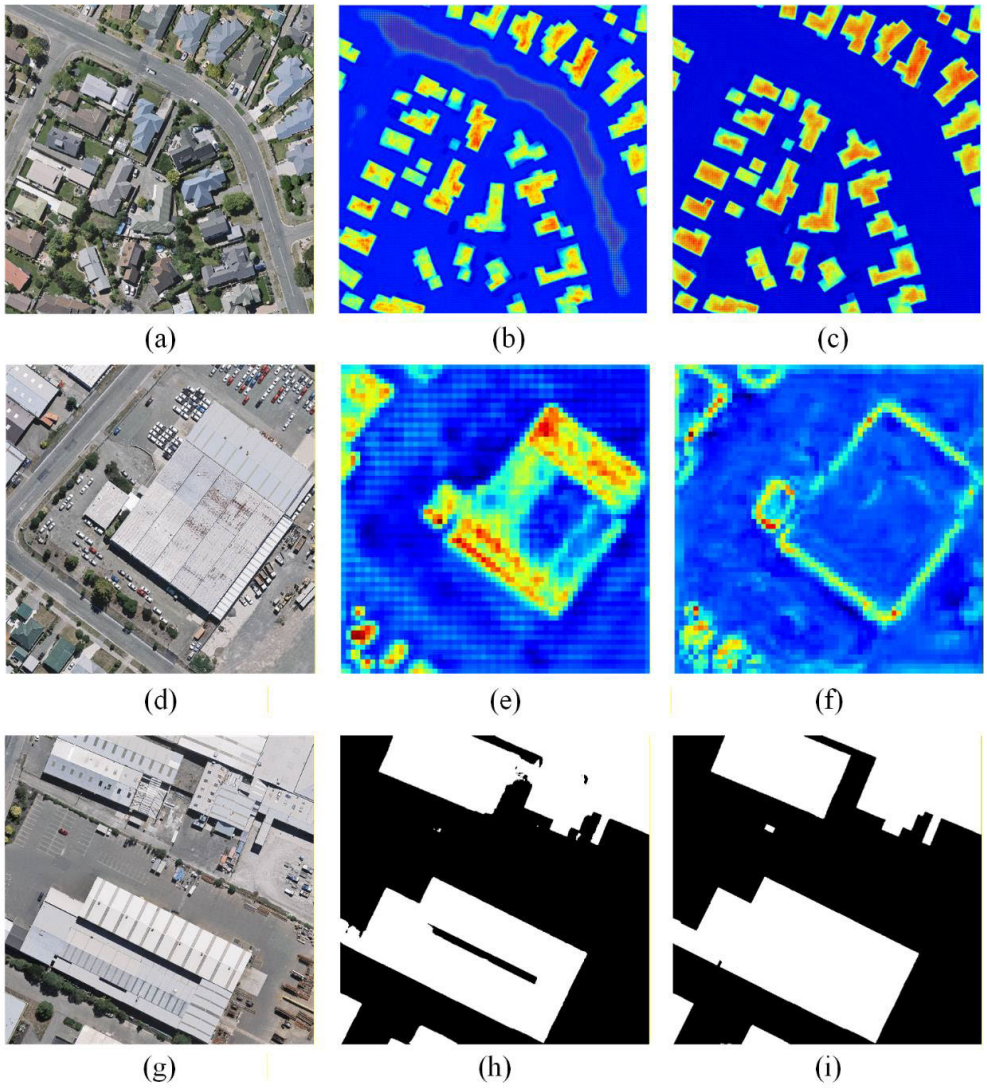


Fig. 7. Visualization of orientation-scale adaptive features and polygon-structure-aware building extraction results. (a), (d), and (g) Input remote sensing images. (b) and (e) Feature responses produced by conventional direction-aware or multiscale mechanisms, which tend to generate scattered activations or emphasize texture-dominant regions. (c) and (f) Feature maps generated by the proposed AMC, where activations are more compact, directionally consistent, and well aligned with dominant rooftop orientations and building structures across different scales. (h) and (i) Compare the final building extraction results without and with the proposed PSE loss, respectively. The PSE module produces more complete and geometrically regular building masks, effectively reducing boundary fragmentation and internal false negatives.

from 91.95% to 91.66%, respectively. On the Inria dataset, the performance degradation is more pronounced, with the $F1$-score decreasing from 90.75% to 90.03% and the IoU dropping from 83.07% to 81.87%. On the Massachusetts dataset, the $F1$-score and IoU decrease from 86.64% to 86.47% and from 76.44% to 76.16%, respectively. These results demonstrate that introducing polygon-level structural constraints during training helps preserve the geometric integrity of buildings, leading to more accurate boundary localization and reduced structural fragmentation, especially in complex urban scenes with dense building distributions and irregular building contours.

Overall, the best performance on all three datasets is achieved when AMC, HSSF, and PSE are jointly integrated, validating the effectiveness of the proposed design. The ablation study confirms that AMC improves orientation-scale adaptability, HSSF enhances spatial-semantic feature alignment, and PSE enforces geometric regularity, and that their combination is crucial for robust and accurate building detection from high-resolution remote sensing imagery.

2) Attention and Feature Visualization Analysis: The feature visualization results are presented in Fig. 7. As shown in Fig. 7(c), the feature maps generated by the proposed AMC module exhibit dense and directionally consistent activations that are well aligned with dominant rooftop orientations, demonstrating strong sensitivity to coupled orientation-scale variations. In contrast, conventional direction-aware mechanisms [Fig. 7(b)] tend to produce scattered or partially misaligned responses under complex spatial layouts.

By jointly predicting optimal kernel rotation angles and scale weights, the AMC module dynamically aligns convolution operations with building geometry, resulting in clearer boundary responses and improved geometric precision. This adaptive behavior is particularly effective for buildings with diverse shapes and orientations within the same scene.

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TABLE II
EVALUATION METRICS OF DIFFERENT ENCODERS ON THE WHU DATASET

Backbone (Encoder)	IoU (%) ↑	F1 (%) ↑	Total Params (M) ↓
ResNet-50	89.91	94.68	45.40
Res2Net-50	91.28	95.44	24.94
ViT-B/16	91.47	95.55	97.94
PVTv2-B2	91.60	95.62	50.84
Swin-T	91.81	95.73	34.85
VGG-16	**91.95**	**95.81**	**22.20**

As illustrated in Fig. 7(f), AMC-enhanced features consistently focus on structurally salient regions across buildings of different sizes. Compared with traditional multiscale mechanisms [Fig. 7(e)], which often emphasize texture-rich yet semantically irrelevant areas, AMC selectively enhances meaningful architectural structures by dynamically adjusting its receptive fields. This leads to reduced redundant activations while preserving essential spatial context.

Finally, the binary building detection results shown in the last row of Fig. 7 indicate that the proposed PSE loss produces more complete and cleaner building masks. By explicitly incorporating polygon-level geometric constraints, the PSE module effectively alleviates boundary fragmentation and internal false negatives, particularly for large industrial buildings, further validating the effectiveness of geometry-aware supervision.

3) Analysis of Different Encoders: To evaluate the compatibility of AMC-PSE-Net with different backbone encoders, experiments are conducted using representative convolutional, transformer-based, and hybrid architectures, including ResNet-50, Res2Net-50, VGG-16, ViT-B/16, PVTv2-B2, and Swin-T. The quantitative results are summarized in Table II.

Overall, VGG-16 achieves the most favorable performance on the WHU dataset. Its densely stacked 3×3 convolutional structure is effective in preserving fine-grained spatial details, which is particularly important for high-resolution remote sensing images containing complex building boundaries and small-scale structures. This characteristic enables AMC-PSE-Net to fully exploit local geometric cues.

Among the convolutional backbones, ResNet-50 exhibits relatively lower accuracy. Although residual connections facilitate deep feature learning, the aggressive downsampling strategy tends to weaken shallow spatial information, which is crucial for precise building delineation. Res2Net-50 improves feature diversity through multiscale residual representations; however, the increased architectural complexity also introduces higher computational cost, which may limit its effectiveness under constrained training conditions.

Transformer-based and hybrid encoders further enhance global contextual modeling and generally deliver competitive performance. However, their advantages come at the cost of increased model complexity and computational overhead. In contrast, VGG-16 offers a more effective balance between feature fidelity and architectural simplicity, making it particularly suitable for the proposed AMC-PSE-Net framework.

In summary, the experimental results indicate that AMC-PSE-Net benefits more from encoders emphasizing stable local feature extraction. The proposed AMC module relies on reliable local spatial representations for orientation–scale adaptation, while the PSE loss requires fine-grained geometric features to enforce effective polygon-level constraints. In this regard, the dense local features provided by VGG-16 are more compatible with the design of AMC and PSE. Although transformer-based backbones offer stronger global modeling capability, VGG-16 achieves better overall effectiveness and robustness when integrated with the proposed modules for high-resolution building detection.

E. Comparison of AMC-PSE-Net With State-of-the-Art Methods

To comprehensively evaluate the effectiveness of the proposed AMC-PSE-Net, extensive comparative experiments are conducted on three public building detection datasets, namely WHU, Inria, and Massachusetts. AMC-PSE-Net is systematically compared with a range of state-of-the-art methods representing different architectural paradigms. Specifically, U-Net [9] is included as a traditional convolutional neural network baseline. SegNet [47], SegFormer [48], and farseg [49] are considered representative semantic segmentation methods. HiSup [25] and HD-Net [26] are adopted as representative polygon-based or boundary-aware building detection and mapping methods. DSAT-Net [11], CSA-Net [50], MDCGA-Net [51], and RDCU-AMSA [52] represent networks that integrate attention mechanisms and deformable structures; and recent transformer-based and hybrid architectures, such as BuildFormer [17] and SDSCUNet [46], are also included for comparison. YOLO-SAM is included as a SAM-based building extraction baseline, where YOLOv8n is used to generate building prompts, and SAM ViT-B is used to produce building masks. Recently proposed methods, including MSGCANet [53] and SegTDformer [54], have also been added to the comparison.

The evaluation is performed using four widely adopted metrics, namely $F1$-score, IoU, precision, and recall, and the overall quantitative results are summarized in Table III and Fig. 8.

To specifically verify the effectiveness of AMC-PSE-Net in geometry-aware building detection, two polygon-based methods, i.e., HiSup and HD-Net, are included in the comparison. Compared with these polygon-based methods, AMC-PSE-Net

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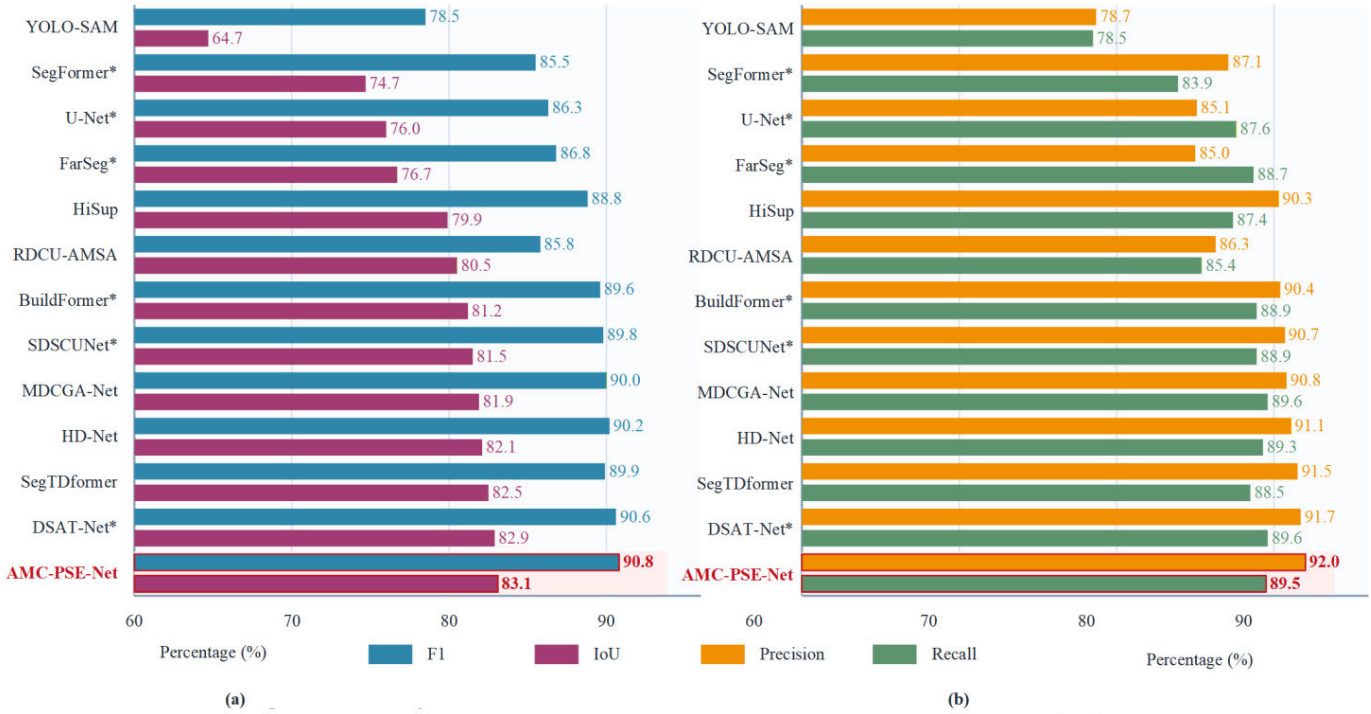


Fig. 8. Quantitative comparison of different methods on the Inria building dataset. (a) Segmentation accuracy comparison in terms of $F1$-score and IoU. (b) Detection quality comparison in terms of precision and recall.

TABLE III

PERFORMANCE COMPARISON TABLE OF DIFFERENT METHODS ("*" DENOTES RESULTS REPRODUCED OR RE-EVALUATED UNDER THE SAME EXPERIMENTAL SETTING, WHILE "—" INDICATES THAT THE CORRESPONDING METRIC IS NOT REPORTED IN THE ORIGINAL ARTICLE)

Method	WHU (%)				Massachusetts (%)			
	F1↑	IoU↑	Pre↑	Recall↑	F1↑	IoU↑	Pre↑	Recall↑
Unet*	94.79	90.09	95.49	94.10	84.74	73.52	86.50	83.05
BuildFormer*	95.37	91.14	95.35	95.38	84.61	73.32	83.78	85.45
SDSCUNet*	94.88	90.27	94.62	95.15	85.66	74.92	87.57	83.84
SegNet	90.96	83.42	89.25	92.87	82.69	70.94	82.83	82.55
SegFormer*	93.51	88.05	94.76	92.30	81.68	69.02	82.48	80.99
FarSeg*	90.79	83.14	87.89	93.89	77.33	63.04	80.54	74.34
HiSup*	91.45	84.25	89.06	93.98	81.53	68.81	83.07	80.04
HD-Net	94.84	90.19	95.00	94.68	86.06	75.53	85.98	**86.13**
CSA-Net	94.60	89.75	95.41	93.80	84.79	73.59	87.27	82.44
DSAT-Net*	95.65	91.66	95.86	95.44	86.62	76.40	87.30	85.94
MSGCANet	95.59	91.53	—	—	86.03	75.47	—	—
SegTDformer	—	—	—	—	84.74	75.47	85.87	83.63
YOLO-SAM*	90.18	82.11	89.00	91.39	77.61	63.41	72.61	83.35
AMC-PSE-Net	**95.81**	**91.95**	**96.10**	**95.52**	**86.64**	**76.44**	**88.02**	85.32

achieves competitive or superior $F1$-score and IoU on the evaluated datasets, demonstrating that the proposed PSE module can effectively exploit polygon-level geometric priors while maintaining robust pixel-level building detection performance.

In addition, YOLO-SAM is evaluated as a prompt-based foundation-model pipeline. It first detects building candidates using YOLOv8n and then uses SAM ViT-B to generate building masks from the detected prompts. Although YOLO-SAM benefits from the general segmentation capability of SAM, its performance is still limited by the quality of detection prompts and the lack of building-specific geometric constraints. As shown in Table III and Fig. 8, YOLO-SAM performs worse than AMC-PSE-Net on all three datasets. For example, on the WHU dataset, YOLO-SAM achieves

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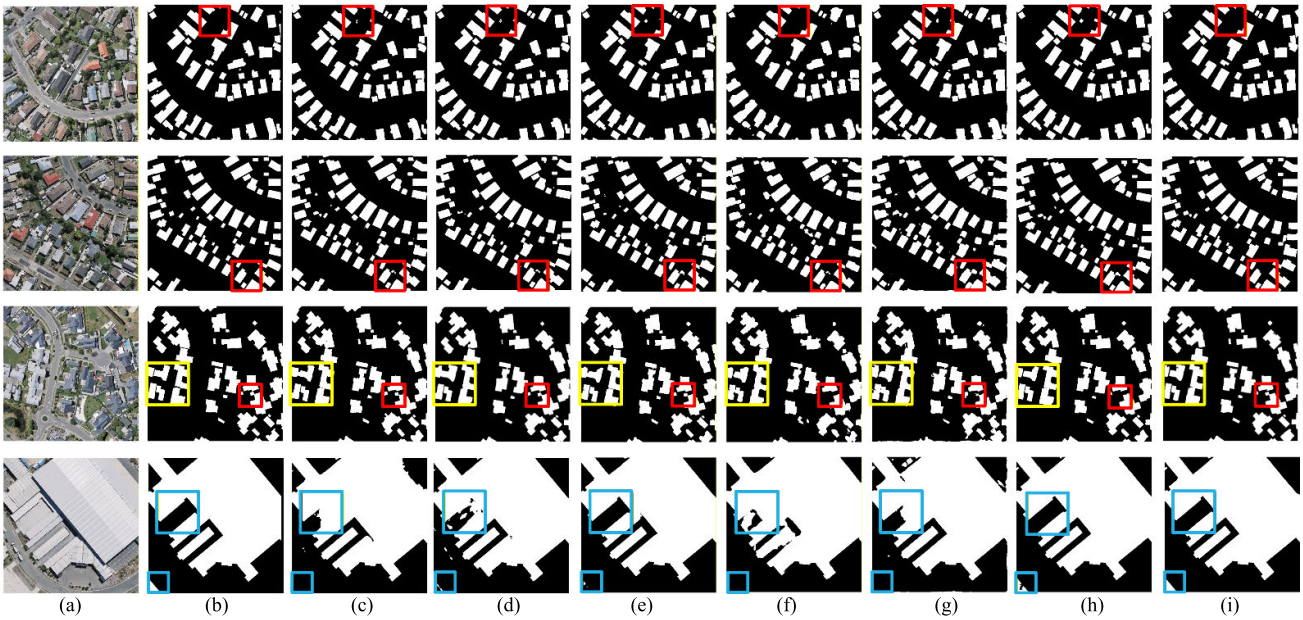


Fig. 9. Qualitative comparison results on the WHU Building dataset. Compared with other methods, AMC-PSE-Net detects small buildings more completely and generates sharper building contours under diverse orientations and complex background interference. (a) Building image. (b) GT. (c) Unet. (d) BuildFormer. (e) SDSCUNet. (f) SegFormer. (g) FarSeg. (h) DSAT-Net. (i) Ours.

90.18% $F1$-score and 82.11% IoU, while AMC-PSE-Net achieves 95.81% $F1$-score and 91.95% IoU. This indicates that task-specific orientation-scale adaptation and polygon-structure regularization remain important for accurate building extraction in high-resolution remote sensing images.

1) Performance Comparison on the WHU Dataset: As reported in Table III, AMC-PSE-Net achieves the best overall performance on the WHU dataset across all evaluation metrics. In particular, AMC-PSE-Net obtains the highest $F1$-score of 95.81%, IoU of 91.95%, and precision of 96.10%, while also maintaining a high recall of 95.52%. Compared with the strong baseline U-Net, AMC-PSE-Net improves $F1$-score and IoU by 1.02% and 1.86%, respectively.

Among the comparison methods, DSAT-Net and BuildFormer show competitive performance due to their spatial attention and global modeling capabilities. The recently proposed MSGCANet also achieves promising results with an $F1$-score of 95.59% and IoU of 91.53%. Nevertheless, AMC-PSE-Net still outperforms DSAT-Net by 0.16% in $F1$-score and 0.29% in IoU, and surpasses MSGCANet by 0.22% in $F1$-score and 0.42% in IoU, indicating its stronger capability in capturing orientation- and scale-variant building structures. These results demonstrate that AMC-PSE-Net achieves a better balance between detection accuracy and boundary delineation.

The qualitative comparison on the WHU dataset is illustrated in Fig. 9. As shown in the first row, the test scenes contain buildings with diverse orientations and small-scale structures. Most comparison methods fail to detect these small buildings, resulting in missed detections or blurred boundaries. In contrast, AMC-PSE-Net accurately segments these buildings with sharp and complete contours.

In the second and third rows, AMC-PSE-Net consistently produces cleaner boundaries and fewer false positives under complex background conditions. This improvement can be attributed to the AMC module, which enhances orientation-scale adaptive feature learning, as well as the PSE mechanism that enforces geometric consistency at the polygon level.

2) Performance Comparison on the Inria Dataset: On the more challenging Inria dataset, AMC-PSE-Net again achieves the best overall performance among all compared methods. As shown in Fig. 8, AMC-PSE-Net reaches an $F1$-score of 90.75% and an IoU of 83.07%, outperforming strong competitors such as DSAT-Net and HD-Net.

Specifically, compared with DSAT-Net, AMC-PSE-Net improves IoU by 0.21% and Precision by 0.28%, while maintaining comparable Recall. Compared with HD-Net, AMC-PSE-Net achieves higher $F1$-score and IoU, demonstrating stronger robustness in complex urban scenes with dense building distributions and diverse background appearances.

Representative visual results on the Inria dataset are shown in Fig. 10. Transformer-based methods, such as BuildFormer and SDSCUNet, tend to produce fragmented predictions or struggle to separate adjacent buildings in dense areas. Attention-based convolutional methods alleviate this issue to some extent but still suffer from boundary irregularities.

In contrast, AMC-PSE-Net effectively distinguishes individual buildings and generates regularized contours, particularly for buildings with complex shapes. The improvement is mainly attributed to the PSE loss, which introduces polygon-level geometric constraints and reduces boundary fragmentation.

3) Performance Comparison on the Massachusetts Dataset: The Massachusetts dataset presents additional challenges due to its lower spatial resolution and complex background interference. Despite these difficulties, AMC-PSE-Net continues to demonstrate superior performance. As shown in Table III, AMC-PSE-Net achieves the highest $F1$-score of 86.64% and IoU of 76.44%, while also obtaining the best Precision of

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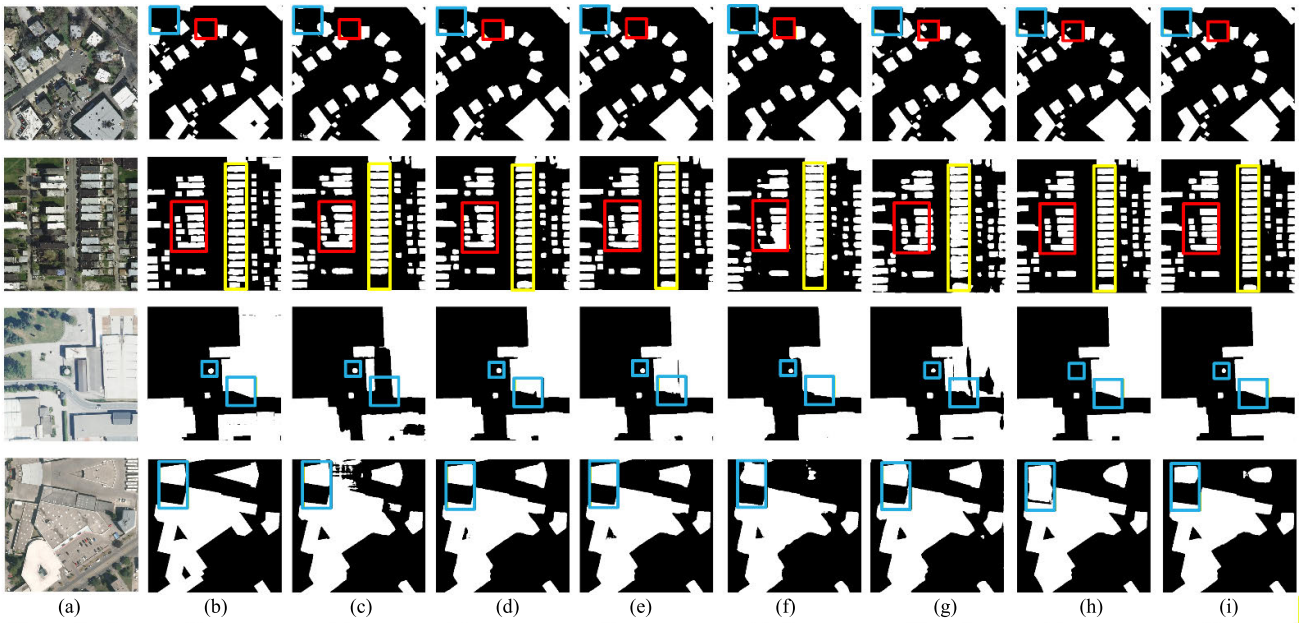


Fig. 10. Qualitative comparison results on the Inria aerial image labeling dataset. AMC-PSE-Net better separates densely distributed adjacent buildings and produces more regular boundaries, demonstrating its robustness in complex urban scenes. (a) Building image. (b) GT. (c) Unet. (d) BuildFormer. (e) SDSCUNet. (f) SegFormer. (g) FarSeg. (h) DSAT-Net. (i) Ours.

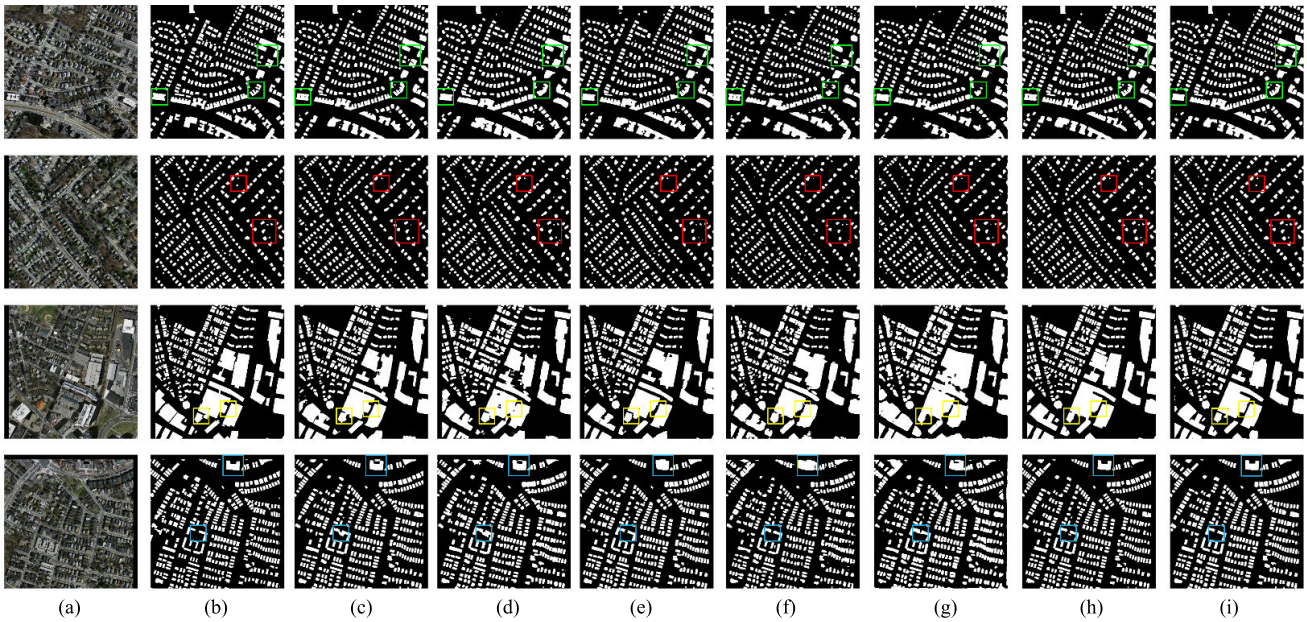


Fig. 11. Qualitative comparison results on the Massachusetts building dataset. The proposed method reduces noisy predictions and boundary fragmentation in low-resolution scenes, resulting in clearer and more geometrically consistent building masks. (a) Building image. (b) GT. (c) Unet. (d) BuildFormer. (e) SDSCUNet. (f) SegFormer. (g) FarSeg. (h) DSAT-Net. (i) Ours.

88.02% among all comparison methods. The recently proposed SegTDformer reports an $F1$-score of 84.74% and IoU of 75.47% on this dataset, which are lower than those of AMC-PSE-Net. MSGCANet yields an $F1$-score of 86.03% and IoU of 75.47%, also falling behind our method.

Compared with the strongest competing method, DSAT-Net, AMC-PSE-Net achieves slight but consistent improvements in $F1$-score and IoU by 0.02% and 0.04%, respectively. More importantly, AMC-PSE-Net obtains a higher precision of 88.02%, outperforming DSAT-Net by 0.72%, although its recall is slightly lower. This indicates that AMC-PSE-Net can reduce false-positive building predictions while maintaining competitive detection completeness. Compared with U-Net and CSA-Net, AMC-PSE-Net shows more pronounced gains in both $F1$-score and IoU, demonstrating its strong generalization capability under low-resolution and complex-background conditions.

The qualitative results in Fig. 11 further confirm these observations. In scenes with densely distributed buildings and complex spatial layouts, most comparison methods produce incomplete or noisy predictions. In contrast, AMC-PSE-Net successfully detects buildings of varying sizes and orienta-

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TABLE IV
PARAMETER USAGE AND FLOPS COMPARISON OF AMC AND RELATED CONVOLUTIONAL VARIANTS ON THE WHU DATASET

Variant	Geometry descriptors	Params (M)	FLOPs (G)	F1 (%)
Baseline	0	21.10	114.63	94.58
Conv	0	22.17	121.64	94.92
Deformable Conv	9 offset vectors	22.24	122.15	95.28
Dynamic Snake Conv	9 constrained offsets	22.36	121.78	95.34
AMC	4 compact descriptor types	**22.20**	**121.73**	**95.81**

tions, generating clear, complete, and geometrically consistent contours. This demonstrates the effectiveness of AMC-PSE-Net in handling challenging building detection scenarios.

F. Parameter Usage and Computational Cost Analysis

To clarify the parameter usage and computational characteristics of the proposed AMC module, we compare it with several commonly used convolutional alternatives under the same controlled framework. All variants are evaluated using 512×512 input images with batch size 1, and the number of learnable parameters, FLOPs, and $F1$-score on the WHU dataset are reported in Table IV.

As shown in Table IV, the baseline model contains 21.10M parameters and 114.63G FLOPs. When AMC is introduced, the number of parameters increases to 22.20M, while the FLOPs increase to 121.73G. Compared with the baseline, AMC introduces only 1.10M additional parameters and 7.10G additional FLOPs. These extra parameters are mainly used to predict compact geometry descriptor types, including the rotation angle, two anisotropic scale factors, and the fusion coefficient, rather than simply increasing model capacity by stacking convolutional branches.

Compared with the standard convolution, deformable convolution, and dynamic snake convolution variants, AMC maintains a comparable computational cost while achieving the highest $F1$-score. Specifically, AMC achieves an $F1$-score of 95.81%, outperforming standard convolution, deformable convolution, and dynamic snake convolution by 0.89%, 0.53%, and 0.47%, respectively. These results indicate that the performance gain of AMC mainly comes from explicit geometry-parameterized kernel adaptation, which is more suitable for buildings with arbitrary orientations, varying scales, and anisotropic structures.

More importantly, the additional parameters in AMC are geometrically meaningful. Standard convolution variants still rely on fixed or predefined receptive-field patterns. Deformable convolution and dynamic snake convolution improve sampling flexibility, but their offsets are not explicitly parameterized according to the orientation and anisotropic scale of building footprints. In contrast, AMC uses a compact set of content-dependent geometric parameters to transform and fuse base kernels in the kernel coordinate space. Therefore, the main advantage of AMC lies in geometry-parameterized feature adaptation for buildings with arbitrary orientations, varying sizes, and anisotropic shapes, rather than in minimizing FLOPs.

These results indicate that AMC introduces a moderate increase in learnable parameters while providing explicit orientation-scale adaptive kernel modeling. The additional parameters are effectively used for building-specific geometric representation, which contributes to improved building extraction accuracy and geometric regularity.

V. CONCLUSION

This article addressed the critical challenges of accurately detecting buildings from high-resolution remote sensing imagery, specifically focusing on the difficulties posed by arbitrary building orientations, scale variations, and the generation of geometrically regular building polygons. We hypothesized that dynamic, geometry-aware feature learning coupled with explicit polygon-level structural supervision would significantly outperform conventional static or pixel-wise methods. Our research culminated in the development of AMC-PSE-Net, a novel end-to-end framework that effectively validates this hypothesis.

The key contributions of this work are twofold. First, the proposed AMC module provides an effective content-adaptive kernel modeling strategy from static kernels to content-adaptive ones, enabling the network to dynamically align its receptive field with diverse building geometries through learned rotation and scaling transformations. Second, the PSE loss advances beyond pixel-level supervision by embedding geometric priors directly into the optimization process. Its EMD-based vertex regression loss robustly handles variable vertex cardinalities, while the self-supervised collinearity-preserving loss effectively suppresses jagged artifacts without compromising sharp corners. Extensive experiments on the WHU, Inria, and Massachusetts building datasets demonstrate that AMC-PSE-Net consistently outperforms state-of-the-art methods, achieving superior $F1$-scores and IoU metrics. These results underscore the framework's effectiveness in producing detection masks that are not only more accurate but also exhibit significantly improved geometric regularity and boundary fidelity, providing a useful geometry-aware solution for high-resolution building extraction for applications in urban planning, population estimation, and detailed cartographic mapping.

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In future work, we will explore the integration of remote sensing foundation models, such as SAM-based segmentation models and vision-language pretraining models, with the proposed polygon-structure enhancement framework to further improve generalization under cross-dataset and low-label scenarios.

Despite the effectiveness of AMC-PSE-Net, several limitations should be further addressed. First, very small buildings may still be missed because their structural cues are easily weakened during downsampling and feature fusion. Second, adjacent buildings in densely distributed urban areas may suffer from adhesion when the interbuilding gaps are narrow or visually ambiguous, leading to merged predictions. Third, the polygon-structure enhancement module depends on reliable boundary and vertex cues, and its performance may degrade under noisy annotations or severe occlusions. In future work, we will investigate small-object-aware feature enhancement, adaptive multiscale supervision, and topology-aware boundary refinement to further improve small building detection and the separation of densely distributed buildings. Moreover, integrating remote sensing foundation models or prompt-guided segmentation models with polygon-structure constraints will be explored to enhance cross-scene generalization and reduce annotation dependence.

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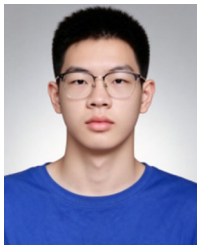
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